Assignment 2

Explore global CO_2 and greenhouse gas data using Python.
Dataset: `CO2_Emissions_Sample`

```
import numpy as np
import pandas as pd
import sklearn
import matplotlib.pyplot as plt
import seaborn as sns
```

Preprocessing

Import

```
# df = pd.read_csv('../assets/CO2_Emissions_Sample.csv')
df = pd.read_csv('https://raw.githubusercontent.com/TheChroniclerr/CP321/refs/heads/main/a2/assets/CO2_Emissions_Sample.csv')
```

```
df.head()
```

	Country	Continent	Year	CO2_Emissions	Methane_Emissions	Nitrous_Emissions	GDP	Population
0	Canada	North America	2000	7750.862683	1053.030592	371.225407	734.472170	3.510898e+07
1	Canada	North America	2001	7955.423607	1023.118633	199.876401	17204.341086	7.396990e+07
2	Canada	North America	2002	8690.632912	994.091186	459.825424	37012.640134	1.296462e+07
3	Canada	North America	2003	NaN	489.361214	460.923247	29251.111335	2.075688e+07
4	Canada	North America	2004	7324.463432	1077.207390	408.154588	23575.082668	1.239332e+07

Clean

```
# Display infos
print(df.isna().sum())
print('------')
print(df.dtypes)
```

```
Country               0
Continent             0
Year                  0
```

```
CO2_Emissions        20
Methane_Emissions    20
Nitrous_Emissions     0
GDP                  20
Population            0
dtype: int64
------
Country               object
Continent             object
Year                   int64
CO2_Emissions        float64
Methane_Emissions    float64
Nitrous_Emissions    float64
GDP                  float64
Population           float64
dtype: object
```

```
# Mean imputate
df[['CO2_Emissions', 'Methane_Emissions', 'GDP']] = (
    df[['CO2_Emissions', 'Methane_Emissions', 'GDP']].fillna(df[['CO2_Emissions', 'Methane_Emissions', 'GDP']].mean())
)
print(df.isna().sum())

# Lock a cleaned copy for export
df_cleaned = df.copy()
```

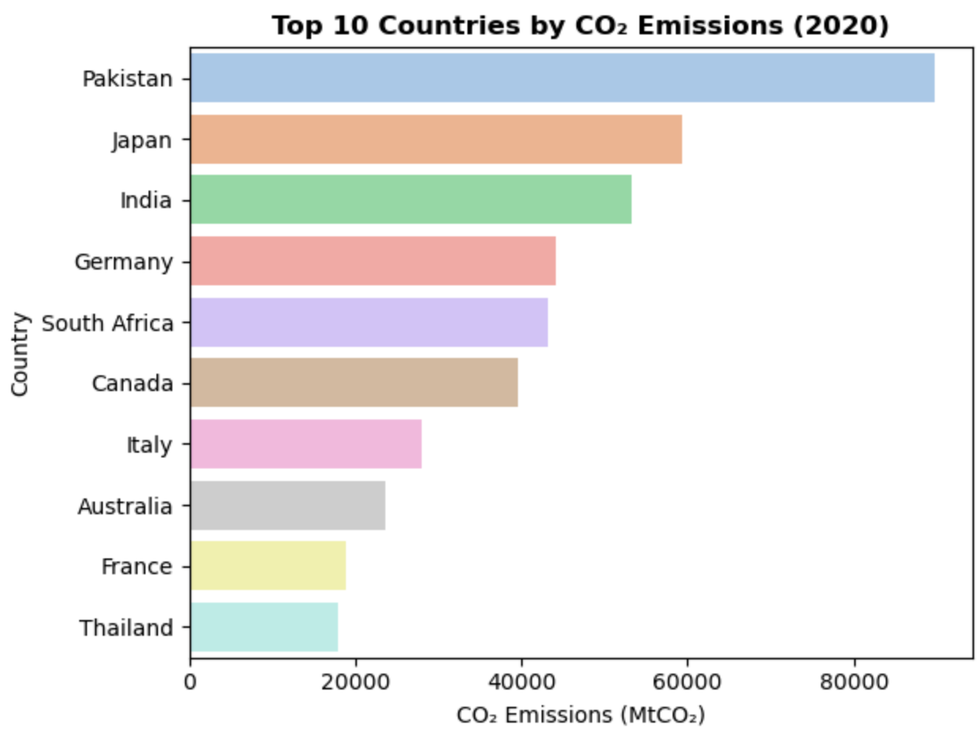
```
Country              0
Continent            0
Year                 0
CO2_Emissions        0
Methane_Emissions    0
Nitrous_Emissions    0
GDP                  0
Population           0
dtype: int64
```

Part A – Visualizing Amounts

Bar Chart

```
df_2020 = df[df['Year'] == 2020]
top10_2020 = df_2020.sort_values('CO2_Emissions', ascending=False).head(10)
```

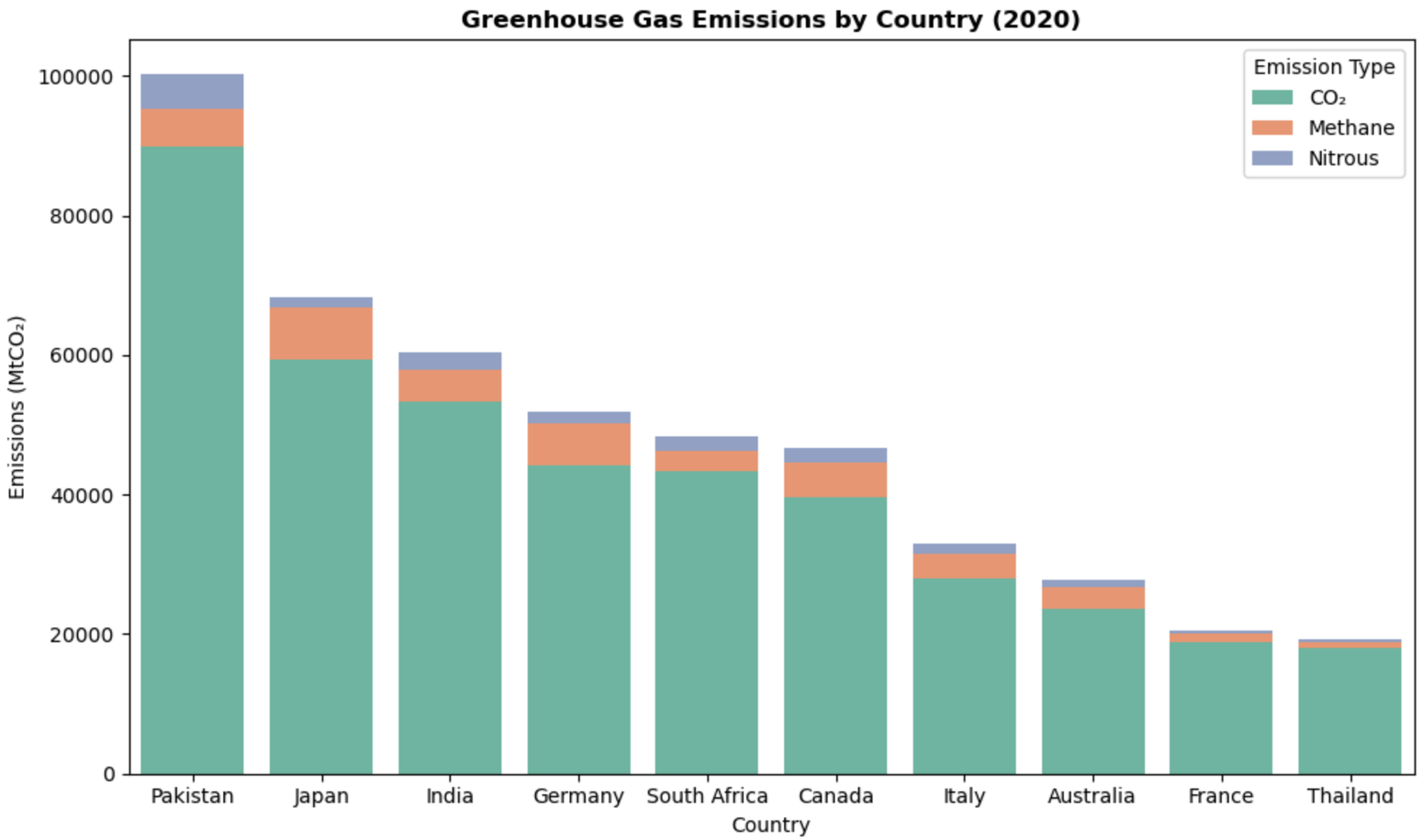
```
plt.figure()
sns.barplot(
    data=top10_2020,
    orient='h',
    x='CO2_Emissions',
    y='Country',
    hue='Country',
    palette='pastel'
)

plt.title("Top 10 Countries by CO₂ Emissions (2020)", fontsize=12, fontweight='bold')
plt.xlabel("CO₂ Emissions (MtCO₂)")
plt.ylabel("Country")
# plt.xticks(rotation=45, ha="right")

plt.tight_layout()
plt.show()
```



Stacked Bar Chart

```
df_long = top10_2020.melt(  # Melt the gas columns into long-form
    id_vars="Country",
```

```python
    value_vars=["CO2_Emissions", "Methane_Emissions", "Nitrous_Emissions"],
    var_name="Gas",
    value_name="Emissions"
)

df_long["Gas"] = df_long["Gas"].replace({   # Rename for cleaner legend
    "CO2_Emissions": "CO₂",
    "Methane_Emissions": "Methane",
    "Nitrous_Emissions": "Nitrous"
})

plt.figure(figsize=(10,6))

bottoms = np.zeros(len(top10_2020))         # Initialize bottom array
for gas in ["CO₂", "Methane", "Nitrous"]:   # Plot each gas type in order
    sns.barplot(
        data=df_long[df_long["Gas"] == gas],
        x="Country",
        y="Emissions",
        bottom=bottoms,  # cumulative bottom
        label=gas,
        color=sns.color_palette("Set2")[["CO₂","Methane","Nitrous"].index(gas)]
    )

    bottoms += df_long[df_long["Gas"] == gas]["Emissions"].to_numpy()   # Update bottoms for next gas

plt.ylabel("Emissions (MtCO₂)")
plt.xlabel("Country")
plt.title("Greenhouse Gas Emissions by Country (2020)", fontsize=12, fontweight='bold')
plt.legend(title="Emission Type")
# plt.xticks(rotation=45, ha="right")

plt.tight_layout()
plt.show()
```

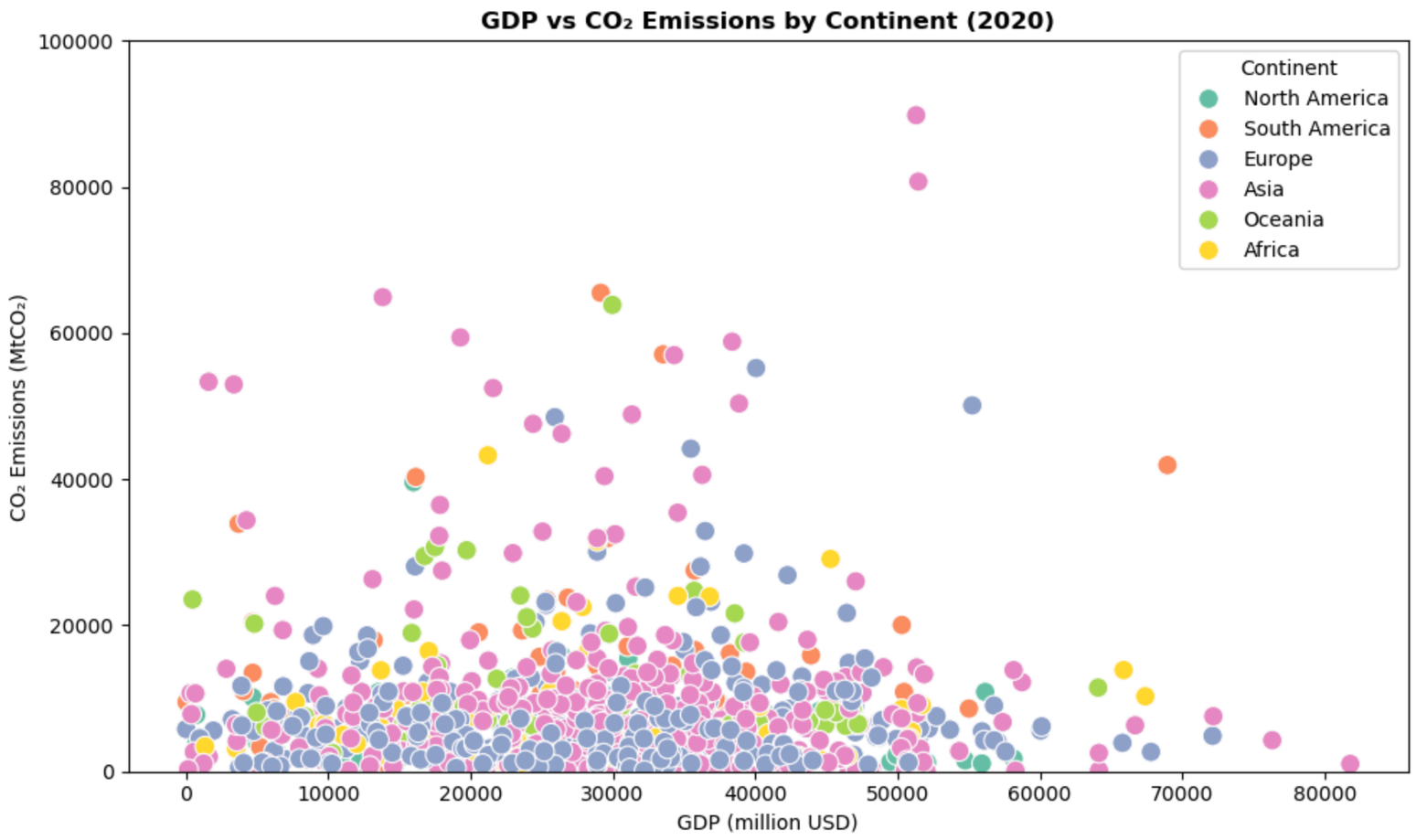


Part B – Visualizing Associations

Scatter Plot

```
plt.figure(figsize=(10,6))
sns.scatterplot(
    data=df,
    x='GDP',
    y='CO2_Emissions',
    hue='Continent',
    palette='Set2',
    s=100   # marker size
)

plt.title("GDP vs CO₂ Emissions by Continent (2020)", fontsize=12, fontweight='bold')
plt.xlabel("GDP (million USD)")
plt.ylabel("CO₂ Emissions (MtCO₂)")
```

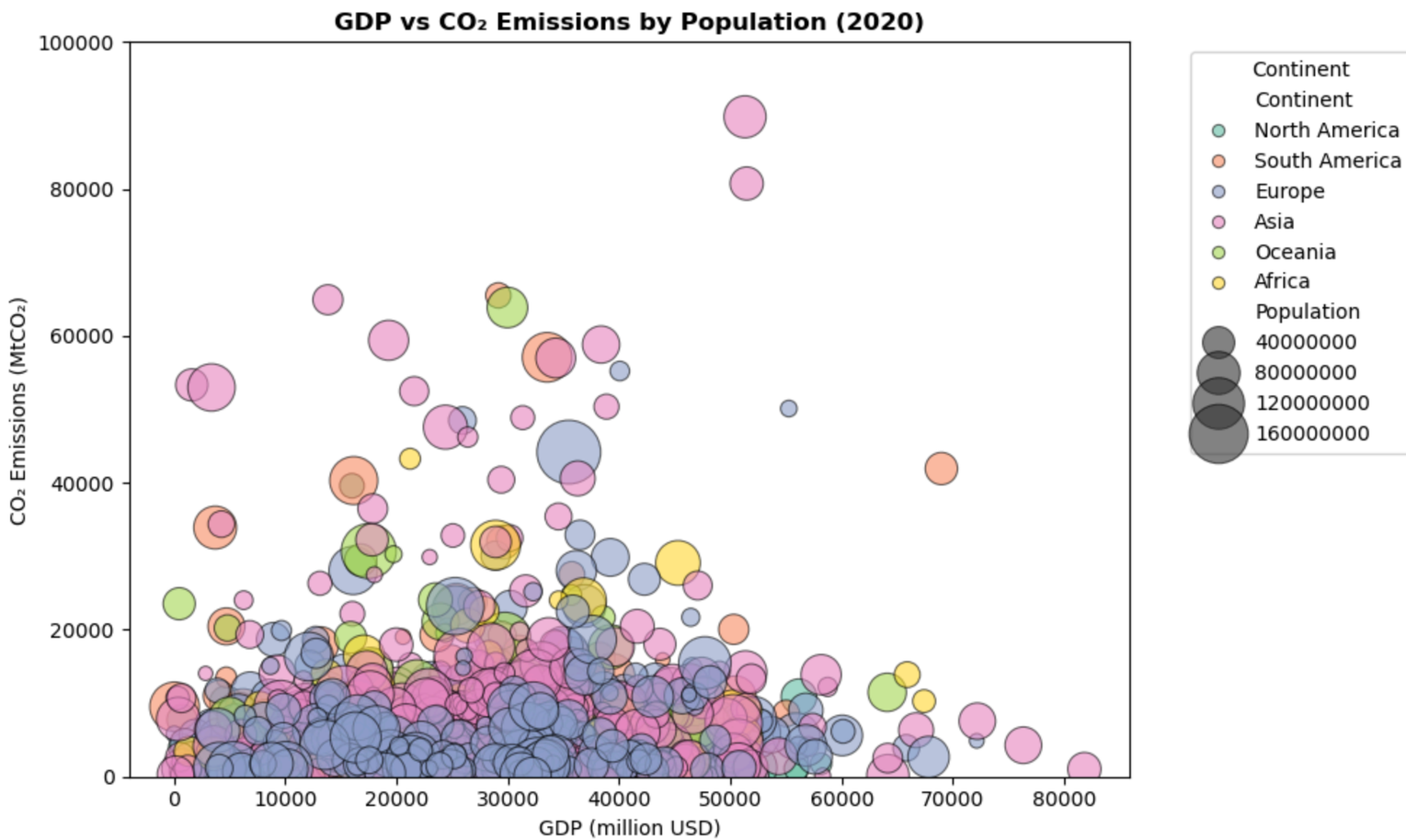
```
# plt.legend(title="Continent", bbox_to_anchor=(1.05, 1), loc='upper left')   # move legend outside the plot

plt.ylim(0, 100000)     # cutoff outliers with >400000
plt.tight_layout()
plt.show()
```



Bubble Chart

```
plt.figure(figsize=(10,6))
sns.scatterplot(
    data=df,
    x='GDP',
    y='CO2_Emissions',
    size='Population',
    hue='Continent',
    palette='Set2',
    sizes=(50, 1000),       # min and max bubble size
```

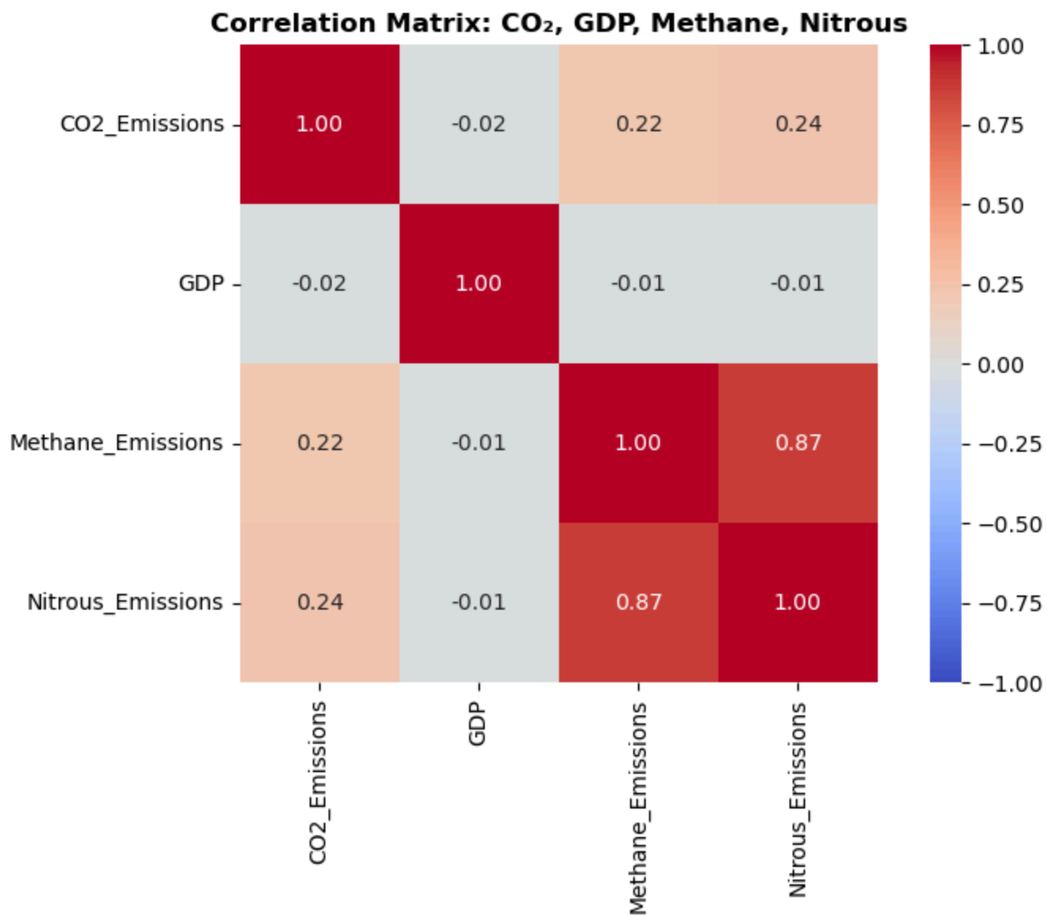
```
    alpha=0.6,              # bubbles transparency
    edgecolor="black"       # bubbles border color
)

plt.title("GDP vs CO₂ Emissions by Population (2020)", fontsize=12, fontweight='bold')
plt.xlabel("GDP (million USD)")
plt.ylabel("CO₂ Emissions (MtCO₂)")
plt.legend(title="Continent", bbox_to_anchor=(1.05, 1), loc='upper left')   # move legend outside the plot

plt.ylim(0, 100000)     # cutoff outliers with >400000
plt.tight_layout()
plt.show()
```



Correlogram

```
df_corr = df[["CO2_Emissions", "GDP", "Methane_Emissions", "Nitrous_Emissions"]].corr()   # compute correlation
```

```python
plt.figure(figsize=(8,6))
sns.heatmap(
    data=df_corr,
    annot=True,       # show correlation values
    fmt=".2f",        # 2 decimal places
    square=True,      # make cells square
    cmap="coolwarm",  # diverging color map: blue-negative, red-positive
    vmin=-1, vmax=1   # set color scale range
)

plt.title("Correlation Matrix: CO₂, GDP, Methane, Nitrous", fontsize=12, fontweight='bold')

plt.tight_layout()
plt.show()
```



Analysis

Outliers

- Most emmision fall <100000, except for 5 outliers >400000.
  - Removed to avoid squashing of the majority dataset.

Scatter Plot

- The plot does not indicate a strong correlation between GDP and emission due to the large spread.
- The scatter plot's legend is ineffective due to large overlaps; however, it is suffient for showing the correlation.

Bubble Chart

- Again, this plot does not indicate a strong correlation.
- The bubble chart legend is better than scatter plot, but it still does not resolve the overlaps and cluttering.

Correlation Matrix

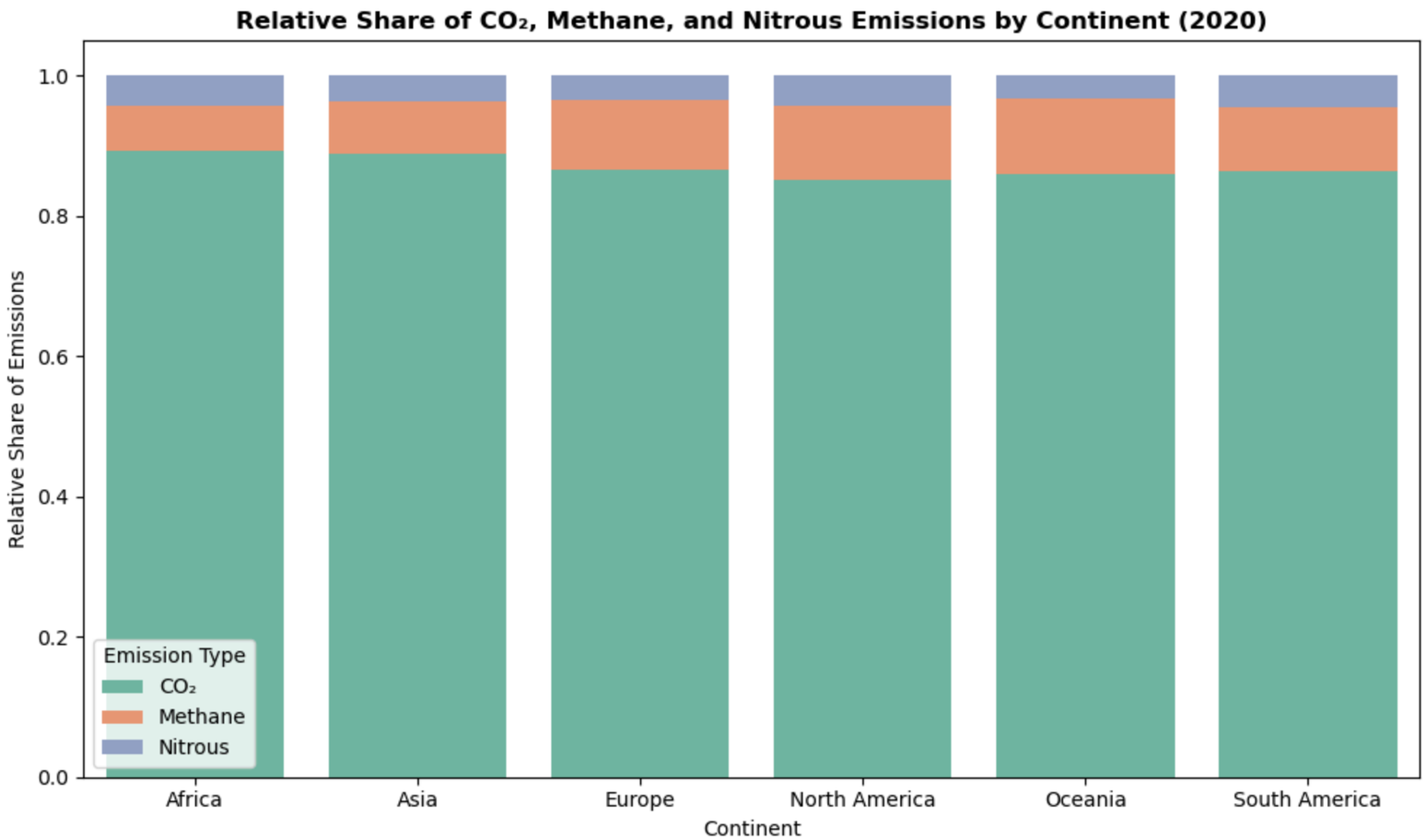
- There is a strong positive correlation between methane and nitrous emissions, indicating that they tend to rise and fall together.
- There is a weak positive correlation between CO_2 and nitrous/methane emissions.
- Correlation matrix is useful for indicating the correlation between each variable.

Part C – Visualizing Proportions

Stacked Bar Chart (Relative)

```
continent_emissions = df_2020.groupby("Continent")[["CO2_Emissions", "Methane_Emissions", "Nitrous_Emissions"]].sum()   # Aggregate emissions by continent
continent_emissions_pct = continent_emissions.div(continent_emissions.sum(axis=1), axis=0)  # Compute relative share (fraction of total emissions per continent)

df_long = continent_emissions_pct.reset_index().melt(   # Melt into long-form
    id_vars="Continent",
    value_vars=["CO2_Emissions", "Methane_Emissions", "Nitrous_Emissions"],
    var_name="Gas",
    value_name="Emissions"
)

df_long["Gas"] = df_long["Gas"].replace({   # Rename variables
    "CO2_Emissions": "CO₂",
    "Methane_Emissions": "Methane",
    "Nitrous_Emissions": "Nitrous"
})

plt.figure(figsize=(10,6))

bottoms = np.zeros(len(continent_emissions_pct))
for gas in ["CO₂", "Methane", "Nitrous"]:
```

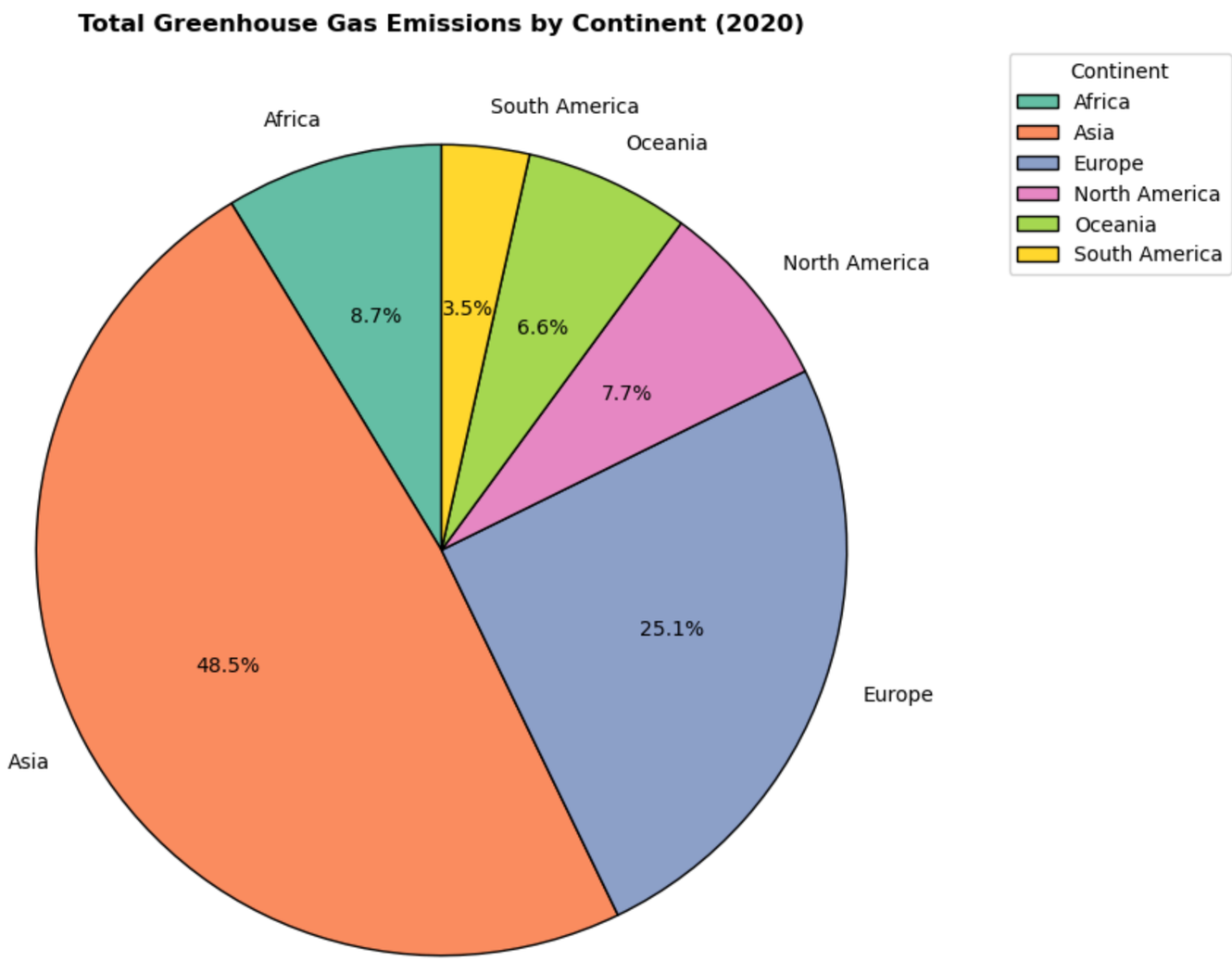
```
    sns.barplot(
        data=df_long[df_long["Gas"] == gas],
        x="Continent",
        y="Emissions",
        bottom=bottoms,
        label=gas,
        color=sns.color_palette("Set2")[["CO₂","Methane","Nitrous"].index(gas)]
    )

    bottoms += df_long[df_long["Gas"] == gas]["Emissions"].to_numpy()   # Update cumulative bottoms

plt.ylabel("Relative Share of Emissions")
plt.xlabel("Continent")
plt.title("Relative Share of CO₂, Methane, and Nitrous Emissions by Continent (2020)", fontsize=12, fontweight='bold')
plt.legend(title="Emission Type")

plt.tight_layout()
plt.show()
```



Pie Chart

```
df_2020_cpy = df_2020.copy()
df_2020_cpy["Total_Emissions"] = df_2020[["CO2_Emissions", "Methane_Emissions", "Nitrous_Emissions"]].sum(axis=1)
region_emissions = df_2020_cpy.groupby("Continent")["Total_Emissions"].sum()

plt.figure(figsize=(9,9))
plt.pie(
    region_emissions,
    labels=list(region_emissions.index),
    autopct="%1.1f%%",      # show percentages
    startangle=90,          # rotate first slice
    colors=sns.color_palette("Set2", len(region_emissions)),
    wedgeprops={"edgecolor": "black"}
)

plt.legend(
```

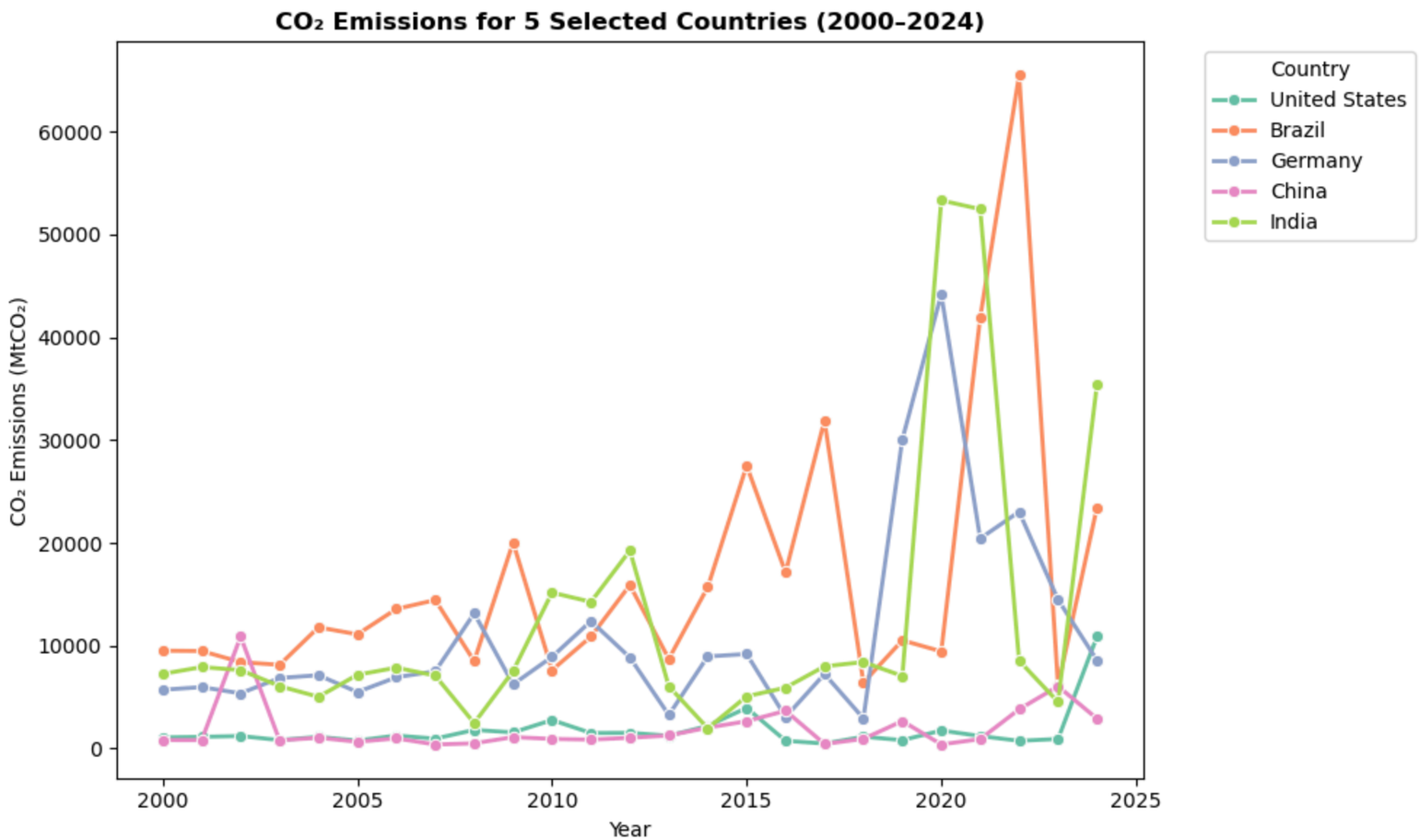
```
    labels=list(region_emissions.index),  # convert Index to list
    title="Continent",
    bbox_to_anchor=(1.05, 1),             # move outside plot
    loc="upper left"
)

plt.title("Total Greenhouse Gas Emissions by Continent (2020)", fontsize=12, fontweight='bold')
plt.show()
```



Part D – Visualizing Time Series

Line Chart

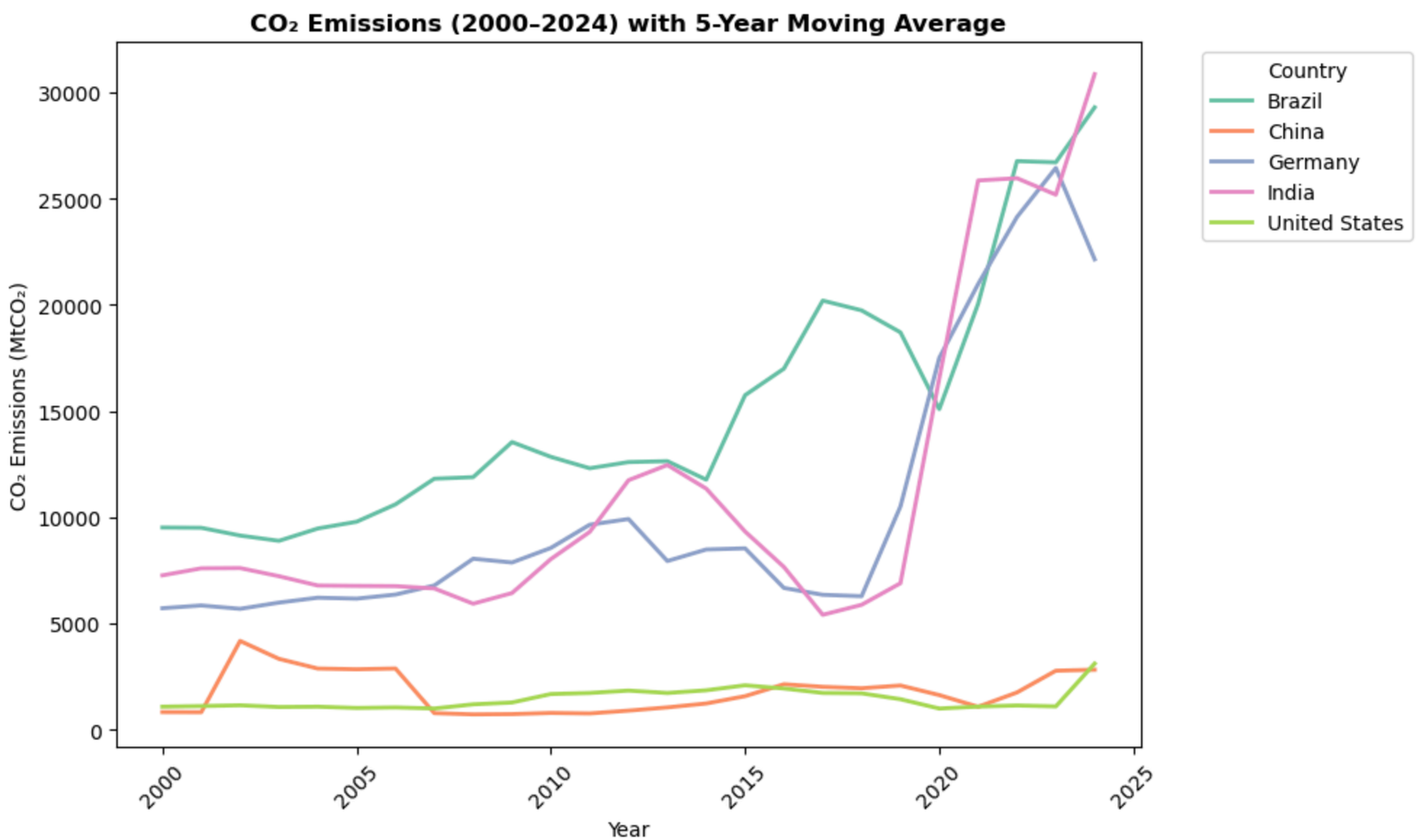
```
selected_countries = ["United States", "China", "India", "Germany", "Brazil"]   # Selected countries
df_selected = df[df["Country"].isin(selected_countries)]

plt.figure(figsize=(10,6))
sns.lineplot(
    data=df_selected,
    x="Year",
    y="CO2_Emissions",
    marker="o",       # markers at each year
    palette="Set2",
    hue="Country",
    linewidth=2
)

plt.title("CO₂ Emissions for 5 Selected Countries (2000-2024)", fontsize=12, fontweight='bold')
plt.ylabel("CO₂ Emissions (MtCO₂)")
plt.xlabel("Year")
plt.legend(title="Country", bbox_to_anchor=(1.05, 1), loc='upper left')
# plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```



Line Chart (Smoothed)

```
df_smoothed = df_selected.sort_values(["Country", "Year"])
df_smoothed["CO2_MA5"] = df_smoothed.groupby("Country")["CO2_Emissions"].transform(lambda x: x.rolling(5, min_periods=1).mean())

plt.figure(figsize=(10,6))

# sns.lineplot(   # Original lines
#     data=df_smoothed,
#     x="Year",
#     y="CO2_Emissions",
#     marker="o",
#     palette="Set2",
#     hue="Country",
#     alpha=0.4,   # lighter so MA stands out
#     linewidth=1
# )
```

```
sns.lineplot(   # Smoothed lines
    data=df_smoothed,
    x="Year",
    y="CO2_MA5",
    hue="Country",
    palette="Set2",
    linewidth=2
)

plt.ylabel("CO₂ Emissions (MtCO₂)")
plt.xlabel("Year")
plt.title("CO₂ Emissions (2000-2024) with 5-Year Moving Average", fontsize=12, fontweight='bold')
plt.xticks(rotation=45)
plt.legend(title="Country", bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```



Analysis

Observation

- The first chart is lot spikier than the second smoothed out chart.
  - The first chart shows more detailed trends per year.
  - The second chart shows more general long term trends.

Discussion

- The **first chart** indicates emission usually spike in certain years instead of being consistently higher, and then lowers back down.
- The **second chart** indicates an increase in emission for developing countries whereas for developed countries its stable/flat.
- Overall, **both charts** indicate developing country in general has much higher emission than developed country.
  - The spikes potentially indicate period of industrialization and development, leading to high emissions.
  - Germany is an exception for it is considered a "developed country". It is still obvious, however, that known developing countries such as Brazil and India shows much higher emissions.

Part E – PCA Representation

1) Data aggregation

```
# a)
print(df_2020.head())

# b)
continent_features = (
    df_2020.groupby('Continent')[[
        "CO2_Emissions",
        "Methane_Emissions",
        "Nitrous_Emissions",
        "GDP",
        "Population"
    ]].mean()
)
print('------')
print(continent_features)
```

```
           Country      Continent  Year  CO2_Emissions  Methane_Emissions  \
20          Canada  North America  2020   39594.595515        5063.162481
45   United States  North America  2020    1751.506594         160.983479
70          Mexico  North America  2020     324.689166          19.082275
95          Brazil  South America  2020    9447.267733        1084.497666
120      Argentina  South America  2020    7266.118286         719.127539
```

```
     Nitrous_Emissions           GDP    Population
20         1927.290993  15995.734800  1.972812e+07
45          102.724595  58152.669169  1.266217e+07
70           15.037827  42698.759788  1.452817e+06
95          435.493745  39287.566755  1.303708e+07
120         431.764424  33624.895797  5.874810e+07
------
               CO2_Emissions  Methane_Emissions  Nitrous_Emissions  \
Continent
Africa          16512.669572        1195.076421         783.345290
Asia            18366.294404        1575.263331         740.888439
Europe           9945.915372        1141.693590         382.081853
North America   13890.263758        1747.742745         681.684471
Oceania         18112.661094        2240.158097         686.367267
South America    6434.510238         681.617460         333.311262

                        GDP    Population
Continent
Africa         31799.308580  1.150217e+07
Asia           29709.777151  4.660265e+07
Europe         29839.928605  6.672596e+07
North America  38949.054586  1.128104e+07
Oceania        11158.775732  7.057577e+07
South America  29192.752776  3.232541e+07
```

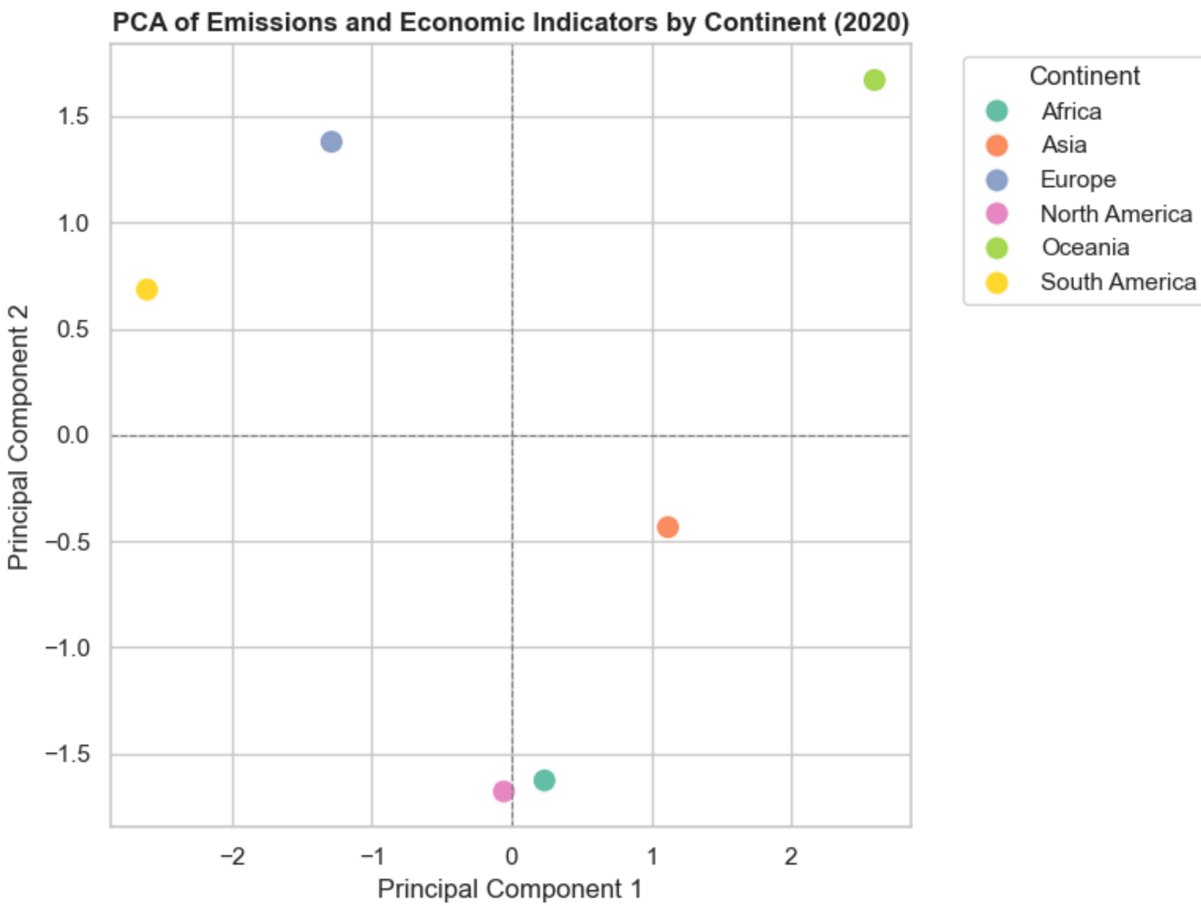
2) PCA feature extraction

```
# a)
scaler = sklearn.preprocessing.StandardScaler()
X_scaled = scaler.fit_transform(continent_features)

# b)
pca = sklearn.decomposition.PCA()
X_pca = pca.fit_transform(X_scaled)

# c)
explained_variance = pca.explained_variance_ratio_ * 100
print(f"PC1 explains {explained_variance[0]:.2f}% of the variance")
print(f"PC2 explains {explained_variance[1]:.2f}% of the variance")
```

```
PC1 explains 55.01% of the variance
PC2 explains 36.02% of the variance
```

3) Visualization

```python
pca_df = continent_features.reset_index()
pca_df["PC1"] = X_pca[:, 0]
pca_df["PC2"] = X_pca[:, 1]

sns.set_theme(style="whitegrid")
plt.figure(figsize=(8,6))

sns.scatterplot(
    data=pca_df,
    x="PC1",
    y="PC2",
    hue="Continent",
    palette="Set2",
    s=120
)

plt.axhline(0, color="gray", linestyle="--", linewidth=0.8)
plt.axvline(0, color="gray", linestyle="--", linewidth=0.8)

plt.title("PCA of Emissions and Economic Indicators by Continent (2020)", fontsize=12, fontweight="bold")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend(title="Continent", bbox_to_anchor=(1.05, 1), loc="upper left")

plt.tight_layout()
plt.show()
```



4) Interpretation

a)
PCA is appropriate after aggregating by continent because the dataset contains multiple correlated variables (CO_2, Methane, Nitrous emissions, GDP, and Population). Aggregation reduces noise at the country level and allows PCA to highlight broad structural differences between continents.

Principal Component 1 (PC1) captures the dominant pattern of variation across continents, which is largely driven by overall scale of emissions and economic size. Principal Component 2 (PC2) captures secondary variation, reflecting differences in emissions composition and economic–population balance that are not explained by PC1.

b)
Continents that appear close together in the PCA plot have similar emissions and economic profiles, while those farther apart differ substantially. Continents with high emissions and large economies tend to score strongly on PC1, while continents with lower emissions and smaller economies appear on the opposite side. Separation along PC2 suggests differences in relative contributions of greenhouse gases or population-adjusted economic characteristics.

Part F – Completeness and Design

✅ Ensure clear legends, color scales, titles, and captions.

✅ Apply the principle of proportional ink.

✅ Avoid clutter and overplotting

Export

```
# Export cleaned CSV
df_cleaned.to_csv("../outputs/cleaned.csv", index=False)
df_cleaned
```

	Country	Continent	Year	CO2_Emissions	Methane_Emissions	Nitrous_Emissions	GDP	Population
0	Canada	North America	2000	7750.862683	1053.030592	371.225407	734.472170	3.510898e+07
1	Canada	North America	2001	7955.423607	1023.118633	199.876401	17204.341086	7.396990e+07
2	Canada	North America	2002	8690.632912	994.091186	459.825424	37012.640134	1.296462e+07
3	Canada	North America	2003	10886.951636	489.361214	460.923247	29251.111335	2.075688e+07
4	Canada	North America	2004	7324.463432	1077.207390	408.154588	23575.082668	1.239332e+07
...	...	...	...	...	...	...	...	...
995	Denmark	Europe	2020	1508.896064	137.853218	77.492468	23858.016712	2.546186e+07
996	Denmark	Europe	2021	1768.923688	161.366640	89.073391	8797.371026	7.945323e+07
997	Denmark	Europe	2022	106.853195	11.279579	5.083564	32222.683168	5.808309e+07
998	Denmark	Europe	2023	8950.158629	951.631105	284.456116	32937.275818	5.557697e+07
999	Denmark	Europe	2024	22483.211561	2137.118754	597.789151	35849.399163	4.378184e+07

1000 rows × 8 columns